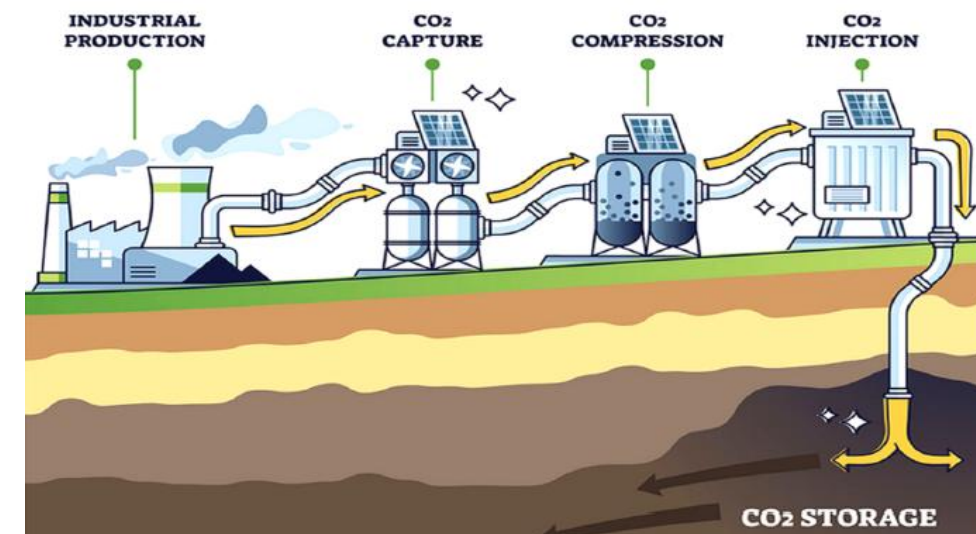
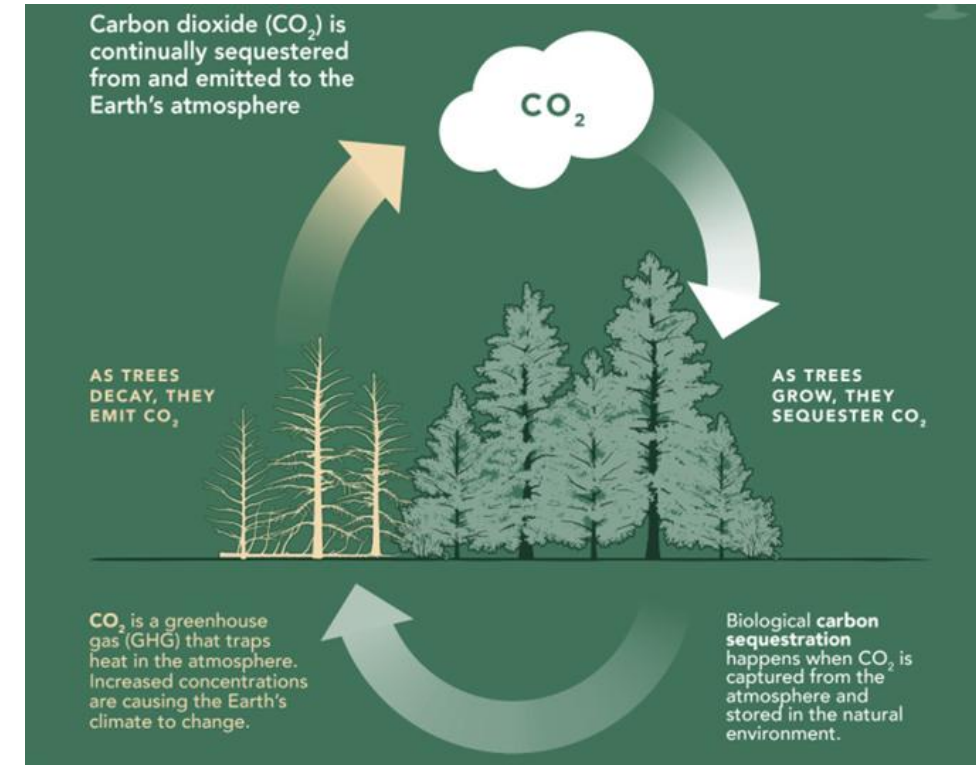


Carbon Sequestration

- **Natural Carbon Sequestration:** Process by which **carbon dioxide (CO₂)** is removed from the atmosphere and stored in natural systems over time.
- **Carbon Capture and Sequestration Technologies (CCUS):** a set of technologies designed to **capture carbon dioxide (CO₂)** emissions from large sources (like power plants or industries), **transport it**, and **store it safely** so it doesn't enter the atmosphere and contribute to climate change.



A close-up photograph of a green leaf with prominent veins resting on the rim of a clear glass test tube. The background is a soft-focus laboratory setting with several other test tubes visible. The text "Natural Carbon Sequestration" is overlaid in white, centered horizontally.

Natural Carbon Sequestration

Major Natural Carbon Sequestration

1. Forests (Terrestrial biomass)

- Trees absorb CO₂ via photosynthesis
- Stored as carbon in trunks, branches, leaves, and roots

2. Soils

- Organic matter (dead plants, microbes) stores carbon
- Grasslands and agricultural soils are also the major reservoirs

3. Oceans

Absorb ~25–30% of global CO₂ emissions. Carbon stored via:

- Dissolution (physical pump)
- Marine organisms (biological pump)

4. Wetlands & Mangroves (Blue Carbon)

- Highly efficient carbon storage per unit area
- Store carbon in waterlogged soils for centuries

Natural systems are generally **in equilibrium** over long timescales.

How forests store carbon?

Forests act as major **carbon sinks** by absorbing carbon dioxide (CO_2) from the atmosphere through **photosynthesis** and storing it in different parts of the ecosystem:

1. Aboveground biomass

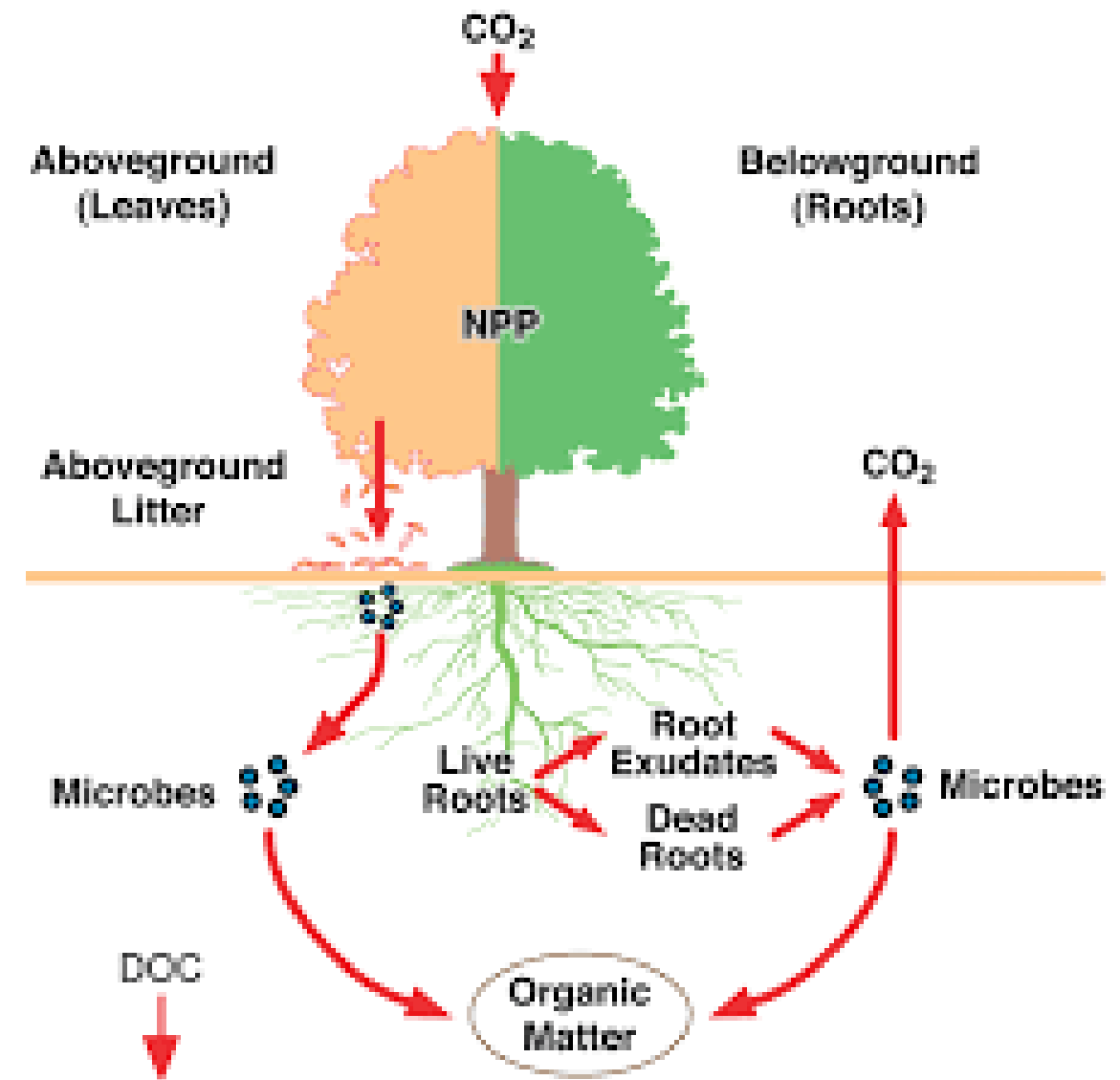
- **Trunks (stems):** Largest carbon reservoir in a tree
- **Branches:** Store structural carbon
- **Leaves:** Temporary carbon storage but crucial for uptake

2. Belowground biomass

- **Roots:** Store carbon and transfer it into the soil

3. Soil organic carbon (indirect storage)

- Fallen leaves, dead roots, and organic matter decompose and enrich the soil
- This creates a **long-term carbon pool**, often larger than the biomass itself



Leaves = carbon capture

Wood (trunk/branches) = carbon storage

Roots & soil = long-term carbon bank

How much do the forests store carbon?

Forest Type	Carbon Stock (tC/ha)	CO ₂ Equivalent (tCO ₂ /ha)
Tropical forests	150 – 300	550 – 1100
Temperate forests	80 – 200	290 – 730
Boreal forests (Taiga) – Conifers (~10%)	50 – 150	180 – 550
Plantations	30 – 150	110 – 550

(1 tonne of carbon (tC) = **3.67 tonnes of CO₂ (tCO₂)**)

Annual Carbon Sequestration Rates

Forest Type	Annual Uptake (tCO ₂ /ha/year)
Natural forests	2 – 10
Fast-growing plantations	8 – 20

(1 hectare of well-managed forest can absorb roughly **5–10 tCO₂/year.**)

India-specific - Average forest carbon stock: **~50–120 tC/ha**, (varies widely). Total forest carbon stock (India): **~7,000+ million tonnes of carbon (FSI)**. Dense forest in the Western Ghats & Northeast can exceed **150 tC/ha**



What happens when you clear the forest?

Forest Type	C_before (tC/ha)	Emissions (tCO ₂ /ha)
Open forest	60	~220
Moderate forest	100	~367
Dense forest	150	~550

Denser forests = much higher emissions when cleared

Agriculture, Forestry, and Other Land Use (AFOLU) is an important sector from Emission point of view

Why Natural Carbon Sink is Important for India: COP 26 (Panchamrit)?

Achieving Climate Goals

- **500 GW** non-fossil electricity capacity by 2030
- **50% energy requirements** from renewables by 2030
- Reduce **1 billion tonnes CO₂ emissions** by 2030
- Reduce emissions intensity by **45%**
- Achieve **Net Zero by 2070**



Non-Fossil Energy Capacity

Achieving 500 GW of non-fossil energy capacity by 2030.



Renewable Energy Source

Sourcing 50% of energy requirements from renewable sources by 2030.



Carbon Emission Reduction

Reducing projected carbon emissions by 1 billion tonnes by 2030.



Economic Carbon Intensity

Lowering carbon intensity of the economy by 45% by 2030.



Net-Zero Goal

Reaching net-zero emissions by 2070.

Agriculture, Forestry, and Other Land Use (AFOLU) is critical for achieving climate goals

Agriculture, Forestry, and Other Land Use (AFOLU)

AFOLU covers **all emissions and removals of greenhouse gases (GHGs)** related to the land use and its management .

1. Agriculture

- Crop cultivation (rice, wheat, etc.)
- Livestock (cattle, sheep → methane emissions)
- Fertilizer use → nitrous oxide (N₂O)

2. Forestry

- Forest growth (carbon sequestration)
- Logging and forest degradation
- Afforestation / reforestation

3. Other Land Use

- Deforestation (forest → agriculture)
- Grasslands, wetlands, peatlands
- Urban land expansion

 Unlike energy, industry or transport sectors, AFLOU is both a **source** *and* a **sink** of emissions

-  Agriculture → emits GHGs
-  Forests → absorb CO₂

AFOLU critical for **net-zero strategies**

Major GHGs in AFOLU

Gas	Source
CO ₂	Deforestation, land-use change
CH ₄ (Methane)	Livestock, rice paddies
N ₂ O	Fertilizers, soils

Note: AFOLU contributes roughly **20–25% of global GHG emissions**, at the same time, forests absorb about **30% of human CO₂ emissions annually**.



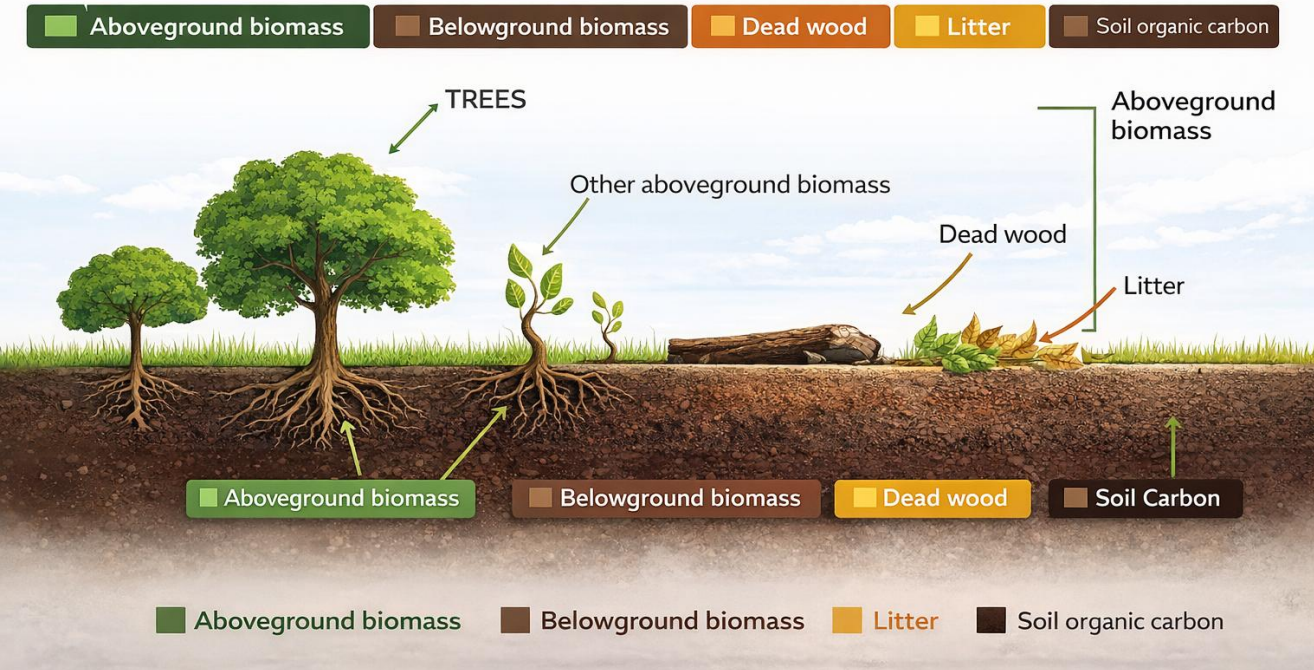
AFOLU Carbon Pool Measurement (IPCC Methodology)

The Intergovernmental Panel on Climate Change (IPCC) provides standardized guidance (2006 Guidelines + 2019 Refinement)

CARBON POOLS (IPCC GPG)

The IPCC GPG (2003) - five carbon pools:

Aboveground biomass, belowground biomass, litter, dead wood, and soil organic carbon



Fundamental Equation

$$\Delta C = C_{t2} - C_{t1}$$

ΔC = change in carbon stock

C_{t1} , C_{t2} = carbon stock at two time points

Converted to CO_2 : $\Delta CO_2 = \Delta C \times 3.67$

Categories:

- Afforestation / Reforestation
- Avoided deforestation
- Forest management

IPCC guidelines for LULUCF: Carbon stock changes in:

- Biomass (trees)
- Soil
- Dead organic matter



Estimation of Aboveground Biomass (AGB)

Allometric equations: $AGB = f(\text{DBH, Height, Wood Density})$

DBH = diameter at breast height (cm)

Example (generic tropical): $AGB = 0.0673 \times (\rho D^2 H)^{0.976}$

D = diameter at breast height (cm)

H = height (m)

ρ = wood density (g/cm^3)

Biomass → Carbon $C_{AGB} = AGB \times CF$

CF (carbon fraction) $\approx$ **0.47** (i.e., **47% of dry biomass is carbon**). Default **value prescribed by IPCC**



Estimation of Belowground Biomass (AGB)

$$BGB = AGB \times R$$

Root-to-shoot ratio (R): ~ 0.2 – 0.3 (default values)

Total Biomass Carbon

$$C_{biomass} = (AGB + BGB) \times CF$$

Stock Change

$$\Delta C = C_{t2} - C_{t1}$$

Soil Organic Carbon (SOC)

$$SOC = BD \times Depth \times C\% \times Area$$

BD (Bulk Density)

Mass of dry soil per unit volume (t/m^3).

- Indicates how compact the soil is.

Depth

Thickness of the soil layer considered (**m**)

- Generally, the topsoil 0–0.3 m, but sometimes up to 1 m.

C% : Fraction of carbon in soil (expressed as %)

- Typically ranges:

0.5–1% (degraded soils)

1–3% (moderate soils)

3% (rich organic soils)

Area: Land area over which SOC is calculated (**m² or ha**)

Dead Organic Matter (DOM)

DOM includes two carbon pools:

Dead wood

- Standing dead trees (snags)
- Fallen logs, coarse woody debris

Litter

- Leaves, twigs, small branches
- Organic debris on the forest floor

$$C_{\text{DOM}} = \text{Biomass}_{\text{DOM}} \times \text{CF} \text{ (t/ha)}$$

Biomass DOM: Total dry mass of dead organic matter

CF (Carbon Fraction): Fraction of biomass that is carbon.

Default $\approx$ **0.47 (IPCC)**

$$\Delta C_{\text{total}} = \Delta C_{\text{biomass}} + \Delta C_{\text{soc}} + \Delta C_{\text{DOM}}$$



AFOLU Exercise

A tropical tree has the following measured parameters:

- Diameter at Breast Height (DBH, D) = 30 cm
- Tree Height (H) = 15 m
- Wood Density (ρ) = 0.60 g/cm³
- Tree Density = 400 trees/ha
- Root-to-shoot ratio (R) = 0.2
- Topsoil = 0.3 m
- Soil carbo = 1%

Using the following generic tropical allometric equation:

$$AGB = 0.0673 \times (\rho D^2 H)^{0.976}, \text{ and}$$

CF (carbon fraction) $\approx$ **0.47 (default IPCC)**

Estimate the total biomass Sequestration Potential. Assume no dead biomass

Forestry Project Exercise:

Parameter	Value
Area	1 hectare
Period	10 years
AGB growth	8 t biomass/ha/year
Root-to-shoot ratio	0.25
Carbon fraction (CF)	0.47
Soil carbon increase	0.5 tC/ha/year
Dead organic matter	0.5 tC/ha (over 10 yrs)

How much 1 ha forest can remove in 10 years? How much Carbon Credit is generated in 10 years?

Living Biomass Change Equation

$$\Delta C_{FSLB} = A \times (C_{after} - C_{before}).....(IPCC)$$

Where:

- ΔC_{FSLB} : Annual Change in the due to conversion of forest to settlement {tonnes Carbon (tC) per year}
- A = area converted (ha/year)
- C_{Before} = carbon stock before conversion (tC/ha)
- C_{After} = carbon stock after conversion (tC/ha)



Living Biomass Change: Exercise

Parameter	Value	Explanation
Area	1 ha/year	Forest cleared per year
C_before	120 tC/ha	Dense forest biomass
C_after	10 tC/ha	Urban vegetation (very low)



India's AFOLU Target

India does **not explicitly label “AFOLU” as a standalone target**, but its AFOLU contribution is embedded in the **forest & land-use (LULUCF) carbon sink goal** under its NDCs which requires an additional **3.5 – 4.0 billion tCO₂e carbon sink** by 2035 through:

- Afforestation: Sink
- Reforestation: Sink
- Agroforestry: Both emission and Sink

(India has already created **~2.29 billion tCO₂e sink (2005–2021)**)

Why This Target is Critical

- Offsets hard-to-abate sectors (steel, cement, transport)
- Reduces pressure on deep decarbonization
- **Co-benefits:**
 - Biodiversity 🌿
 - Water security 💧
 - Rural livelihoods 👨🌾



Exercise

How much forest area is needed to meet India's **forest & land-use (LULUCF) carbon sink goal** additional **3.0 billion tCO₂e carbon sink** by 2030 at an average forest sequestration value of **6 tCO₂/ha/year**?

Oceans Carbon Sequestration

Oceans play a critical role in regulating the global climate by absorbing approximately **25–30% of anthropogenic CO₂ emissions** each year. This makes them one of the largest natural carbon sinks on Earth. Oceanic carbon sequestration occurs primarily through two key mechanisms: the **physical (solubility) pump** and the **biological pump**.

 **Physical Pump (Solubility Pump):** Direct absorption of CO₂ from the atmosphere into seawater.

Key Processes:

- CO₂ gets dissolved in surface ocean waters based on the differential partial pressure between atmosphere and ocean.
- **Temperature:** Cold waters (e.g., polar regions) absorb more CO₂ than warm waters.
- Dissolved CO₂ reacts with water to form: $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{HCO}_3^- \rightleftharpoons \text{CO}_3^{--}$
 - **~90%+ of carbon is stored as bicarbonate (HCO₃⁻)**
 - Only a **small fraction remains as dissolved CO₂**
- **Buffering by Ocean Chemistry (Carbonate System):** The ocean has a powerful **chemical buffering system**.
 - If CO₂ tries to increase → shifts equilibrium toward bicarbonate
 - If CO₂ decreases → shifts back

Oceans Carbon Sequestration

- **Deep Ocean Transport:** Once surface water sinks, It enters the **deep ocean (1000–4000 m)**, becomes part of **thermohaline circulation**. Through **thermohaline circulation**, carbon-rich surface waters sink and transport CO₂ into the deep ocean, where it can remain stored for **hundreds to thousands of years**.
- **Timescales:** Deep water circulation: **hundreds to ~1000 years**, Some water masses: **>2000 years**. Carbon is **physically trapped away from the atmosphere**. It **cannot exchange gases** until it resurfaces.
- When deep water eventually returns to warmer regions, CO₂ *can* be released. But:
 - Upwelling is **slow and geographically limited**
 - Release is **gradual, not instantaneous**
- Much carbon is **reused biologically before degassing (Biological Pump)**. **Biological Pump Locks Additional Carbon**
Before resurfacing:
 - Some carbon is converted into **organic matter**
 - Sinks further or gets buried in sediments

This creates **longer-term (centuries–millennia) storage**

Oceans Carbon Sequestration

Biological Pump

The **biological pump** involves the uptake of CO₂ by marine organisms and its transfer to deeper ocean layers.

Key Processes:

- **Photosynthesis by Phytoplankton:**

Phytoplankton absorb CO₂ and convert it into organic carbon using sunlight.

- **Food Web Transfer:**

Carbon moves through marine food chains (zooplankton → fish → higher trophic levels).

- **Sinking of Organic Matter:**

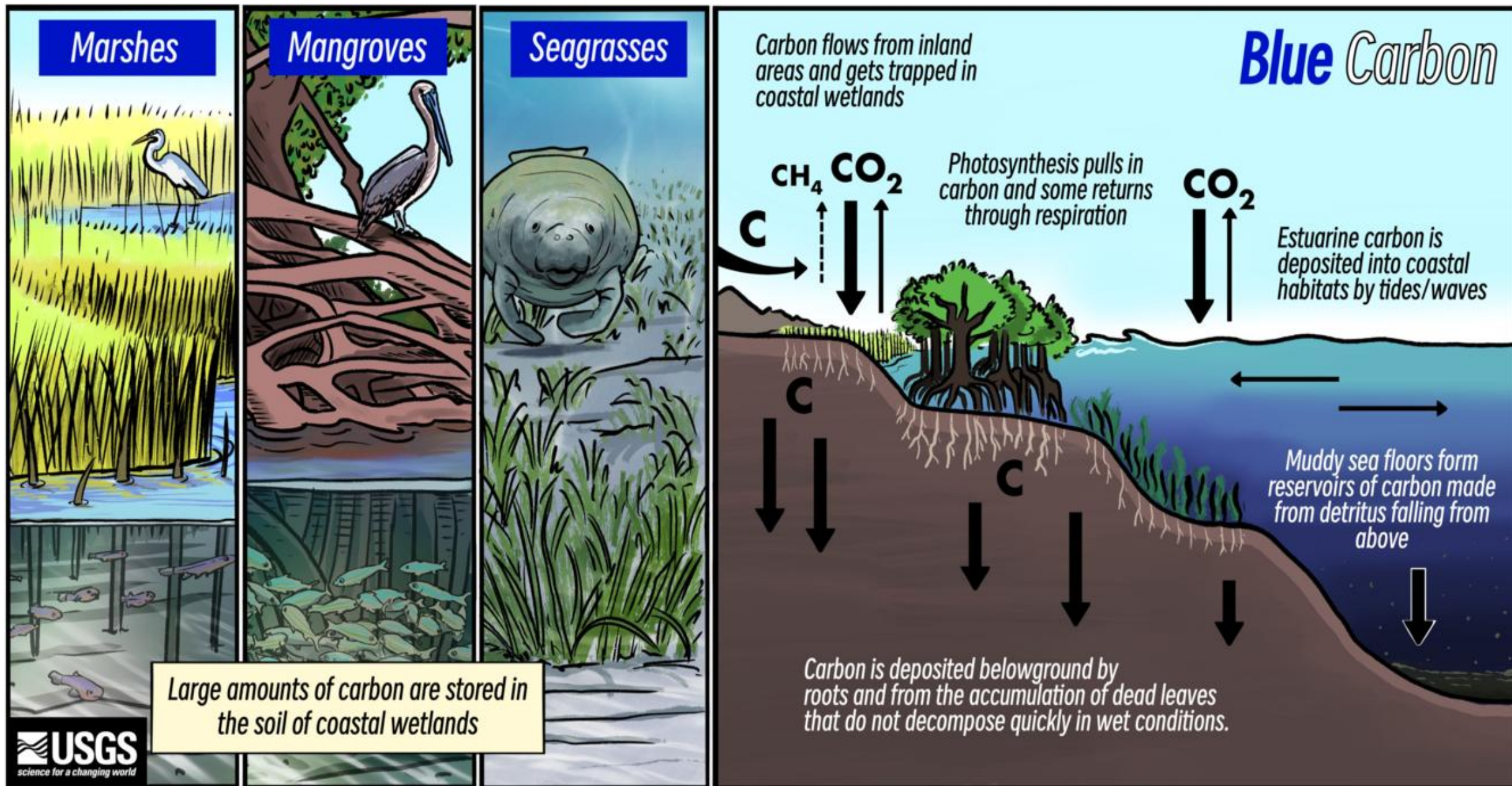
Dead organisms, fecal pellets, and organic debris sink to the deep ocean (often called **marine snow**).

- **Carbon Burial:**

A fraction of this carbon is buried in ocean sediments, effectively removing it from the atmosphere for **millennia to geological timescales**.



Wetlands & Mangroves Sequestration (Blue Carbon)



Blue Carbon?

Blue carbon refers to the carbon captured and stored by **coastal and marine ecosystems**, mainly:

- **Mangroves** (~6–8 tCO₂/ha/year)
- **Salt marshes** (**Salt marshes**: ~4–6 tCO₂/ha/year)
- **Seagrass meadows**: (~2–4 tCO₂/ha/year)

These ecosystems are among the **most efficient natural carbon sinks on Earth**, storing carbon both in vegetation and deep sediments (Up to **1,000+ tCO₂ per hectare** stored in soils).

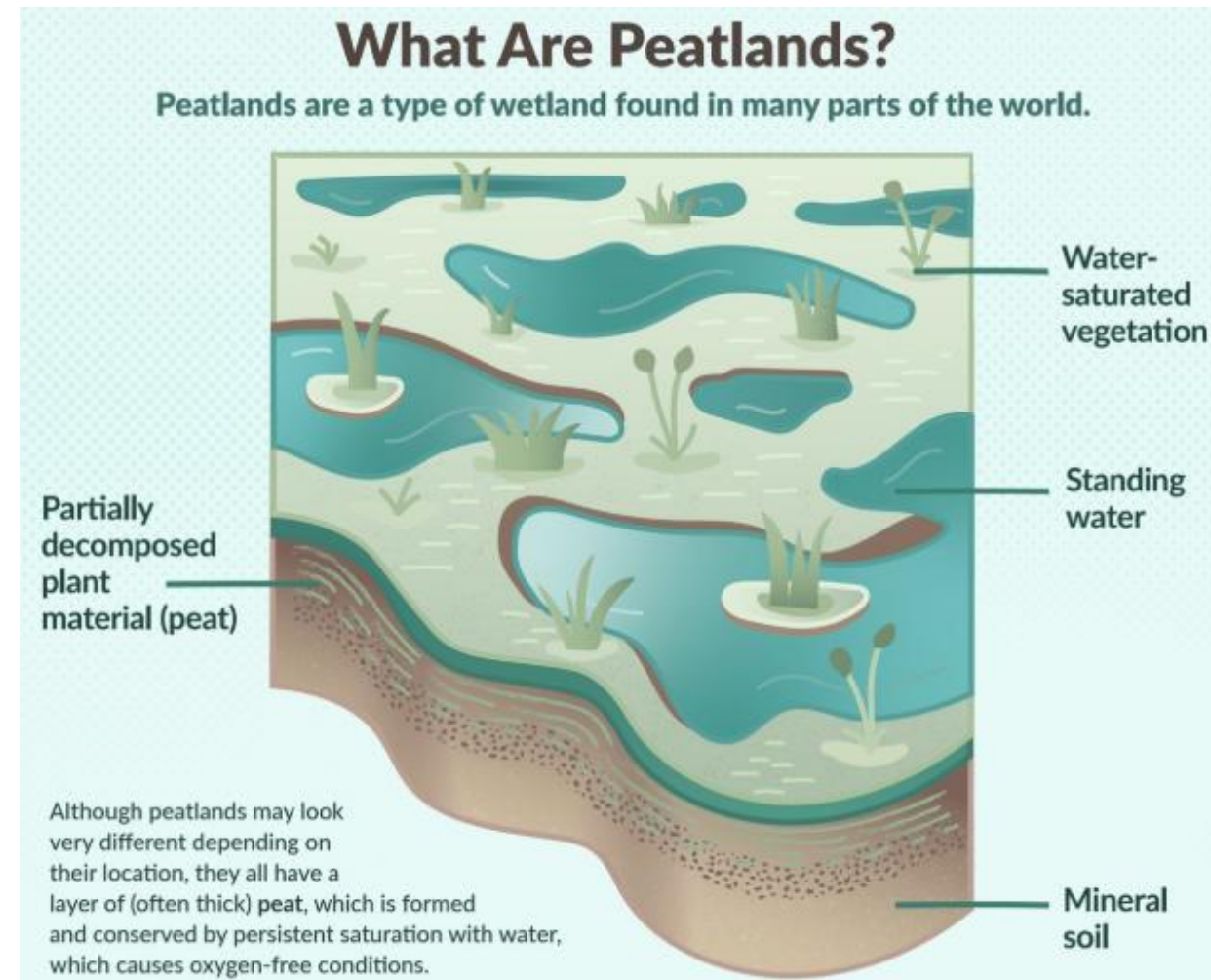
Sequestration Process

1. **Photosynthesis (Carbon Uptake)**: Similar to forests, but often **faster per unit area**,
2. **Belowground Storage (Key Advantage)**: A large fraction of carbon is stored in: Roots, Organic-rich soils (sediments).
Waterlogged (low-oxygen) conditions cause **slow decomposition**. Carbon accumulates over **centuries to millennia**
3. **Sediment Trapping**: Mangrove roots and wetland vegetation trap sediments carried by water and lock organic carbon into soils, creating **deep carbon-rich layers** (up to several meters, unlike forest soil layers which are relatively thin).
4. **Long-Term Burial**: Unlike terrestrial forests (where carbon can re-enter atmosphere quickly), blue carbon ecosystems:
 - Store carbon in **stable, anaerobic sediments**
 - Have **low disturbance rates (if protected)**

Peatlands Carbon Sequestration

Peatlands are waterlogged ecosystems (bogs, fens, and peat swamps) where partially decomposed plant material accumulates as **peat soil** over thousands of years.

- Found in regions like **Indonesia, Congo Basin, Canada, and Northern Europe**
- Cover only **~3% of Earth's land surface**. ~ 25% of the global soil carbon stock—twice the amount of world's forests
- Store **~550 billion tonnes of carbon** — more than all the world's forests combined.





3% of Land Surface



550 Billion
Tonnes of Carbon Stored



Peat Depth
>10 Meters

CO₂ Photosynthesis



Plants Absorb CO₂

Sphagnum Moss



Slow Decomposition
Waterlogged & Low Oxygen
Carbon Accumulates



Plant Litter

Peat Layer

Deep Peat Deposits

Carbon Accumulation
= 0.5–1 mm/year

Carbon Storage
> 500–2,000 tC/ha

Stored Carbon
> Twice as Much as
All the World's Forests

 **Peatlands Carbon Sequestration Process**

Threats:
Drainage & Fires



Restoration:
Rewetting & Rewilding



Peatlands Carbon Sequestration

Sequestration Process

1. Continuous Carbon Accumulation

- Plants absorb CO₂ via photosynthesis
- Dead biomass accumulates as peat layers ($\approx 0.5\text{--}1$ mm/year).

2. Slow Decomposition:

- Waterlogged (anaerobic) conditions limit microbial activity
- Plant material does not fully decay- carbon stays locked in soil

3. Long-Term Storage

Carbon is stored for **thousands of years**

Peat depth can exceed **10 meters**

Benefits of Healthy Peatlands

Peatlands provide important
ecosystem services and support
people and culture worldwide



Impact of Degraded Peatlands

The degradation of peatlands, such as drainage can have far reaching impacts

